

Architecting a Profession of Enterprise Architecture

Most true professions evolve to that state over many decades, or even centuries. We know them when we see them, but is it possible to accelerate the process? What makes a line of work become a profession?

This is a question that the IEEE Computer Society has previously addressed in the formation and recognition of Software Engineering as a profession. As a founding member of the Federation of Enterprise Architecture Professional Organizations (FEAPO), the Computer Society is sharing its experience and its model of a profession with FEAPO. Significantly, the Center for the Advancement of the Enterprise Architecture Profession (CAEAP), another founding member of FEAPO, had also done quite a bit of work in defining Enterprise Architecture as a profession. Both organizations had identified similar hallmarks of a profession. At the same time, Penn State University was addressing the need for curricula to educate students about Enterprise Architecture.

Why do we need a profession of Enterprise Architecture?

The most critical issue facing Government, Defense, Commercial and non-profit enterprises today is the rapid pace of change in almost every area of organized endeavor. With the rate of technological change increasing, together with today's budget and competitive pressures, enterprises must be able to change rapidly often just to survive - let alone succeed. More and more organizations are turning to the evolving science of Enterprise Architecture to develop solutions to these challenges.

Increasing the amount of computer hardware, software, packages, staff, or outsourcing is not the solution; it is a large part of the problem. Rather, the solution requires strategic direction from senior management and strategic planners, with these directions then translated into rapid action by professionals who possess a solid understanding of both enterprise technology and the enterprise. This is the realm of Enterprise Architecture.

Enterprise architecture enables the design and implementation of the structures that link an organization's strategy with its execution. This vital link captures the organizational strategy as blueprints that include enough guidance and detail for the various parts of the organization to execute while facilitating collaboration and innovation. Enterprise Architects use specialized practices to determine where the company is today, scenarios for where it will be tomorrow, and they provide roadmaps that lead from one stage in the journey to the next.

EA advises on investments, opportunities, and risks:

- **ORGANIZATIONAL CONTEXT**

Impacting the formal structure, work management practices, human resource policies, leadership, and culture

- **CORPORATE VISION**

Mapping and alignment with the mission, objectives, goals, tactics, and strategies

Commented [jw1]: Be careful about inserting IT focus into the justification at this point, lest we relegate EA as another IT-derivation. EA is much more than the IT stuff and we need to focus on the business end of the problem (if you will). The core problem with many organizations is the increasing focus on tactical and short-term solutions without the understanding of the complexity of unintended consequences caused by uninformed decisions. This is where EA brings a more holistic understanding of the relationships across the myriad of structures w/in and w/out a given organization. The key is how that complexity is communicated in a way that the decision maker has a better understanding of the context within which to make a more informed decision (didn't say the right decision...), which is hopefully a better one. This is the way EA can enable sustainability for an organization and the stakeholders that it serves.

- **VALUE CHAINS**

Strengthening alliances, capabilities, services, and processes by determining the return on technology investments.

- **Planning**

Rationalizing roadmap investments with respect to capabilities, resources, and competencies.

- **Competitive Forces**

Recognizing internal and external competitive forces that influence product and service offerings and their relationships.

- **Common Language**

Establishing and maintaining the vocabulary, taxonomies, concepts, patterns and references used to frame the organization.

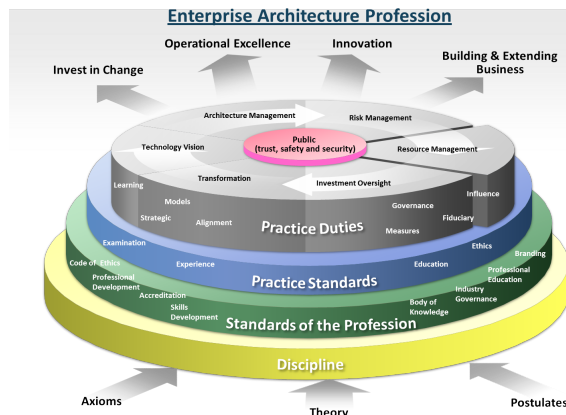


Figure 1: The Enterprise Architecture Profession

What are the issues and what can we do to fix them?

The legitimacy and formalization of EA is not consistent across organizations. This results from a lack of agreed knowledge about EA. As a result, views of EA are typically constrained by political, cultural, and social elements instead of professional status. EA does not yet have the professional standing found in well-established professions like medicine, law, engineering, and accounting. Rather, as in the early days of other, older professions, there are many grass-roots organizations that have sprung up, as practitioner themselves

EXAMPLE PRINCIPLES for the Profession of Enterprise Architecture*

Serve Society Today

Enterprises, governments, and individuals have immediate and pressing need for the duties and advice of the EA Profession.

Adapt for Tomorrow

Society's needs are diverse and continually evolving; the EA Profession responds to, reflects, and anticipates these changes in tomorrows' needs.

Establish and Uphold Ethical and Professional Standards

Integrity is the bedrock on which trust and respect are built; the integrity of the EA Profession is measured by the soundness of its professional practices.

Develop an Empirical Body of Knowledge

A recognized, empirically derived body of knowledge forms the basis for EA practices, and ensures that members of the profession have a common understanding of these practices.

Demonstrate Self-Regulation

Professional autonomy is earned through the strict enforcement of ethical and professional standards; establishment of accreditation models, certification requirements, and professional registries; and through clear and compelling leadership.

*Courtesy of CAEAP

Commented [jw2]: I think the issues with EA is that there is no baseline of knowledge nor are there standards and guidelines to ensure consistency of service delivery w/in the context of a profession. Formalization comes after standards are published and agreed upon by the body of practitioners. The practice of the profession also requires a governing body to assure practitioners are held to the agreed upon standards (and ethics) to gain legitimacy as a profession. Whether EA is adopted by organizations as part of their operational or business models is a test of the legitimacy of the profession. This is where case history (evidence-based results) would be useful to capture and communicate if possible.

have struggled to get agreement about EA practices, EA artifacts, and EA benefits to the enterprise.

Prior to FEAPO, these fragmented efforts have failed to coalesce and to set out on a path to sustained legitimacy and a clear direction in pursuit of professionalism. There is no commonly agreed definition of a clear and consistent set of Enterprise Architect capabilities or a clear differentiator between EA and the many flavors of “Architect” in Enterprise IT.

FEAPO’s goal is to guide the industry to a self-governed status of maturity by 2015 - moving from organized, to qualified, to self-governed which includes a measured level of maturity in three very distinct broad areas:

1. Standards of practice
2. Professional learning
3. Industry self-governance and concomitant professional autonomy

As part of this effort, FEAPO will:

1. Clarify the professional Enterprise Architect’s contributions to the enterprise and establish brand recognition for the profession
2. Assure the public they’re dealing with competent EA professionals, enabling the public’s trust in EA as a profession
3. Provide for sustainability of the profession

Convergence

The Federation of Enterprise Architecture Professional Organizations (FEAPO) is a worldwide association of professional organizations which have come together to provide a forum to standardize, professionalize, and otherwise advance the profession of Enterprise Architecture. It provides a focused hub for the Enterprise Architecture expertise and resources of its member organizations, which include, in addition to the IEEE Computer Society and the Center for the Advancement of the Enterprise Architecture Profession, such organizations as The Association of Business Process Management Professionals, The Business Architecture Guild, the Business Architecture Society, the Canadian Information Processing Society, Data Management International (DAMA), and the International Council on Systems Engineering (INCOSE) .

Commented [jw3]: Wondering if it would be helpful to soften the approach and suggest that these established organizations are attempting to guide alignment of a wide set of players across multiple domains to move the “collective” towards a more consistent level of understanding (BoK), consistent practice of the profession (standards), and build a model for ongoing collaboration to drive continuous improvement (Sustainability).

The Road Ahead

Some practitioners do not see a need to professionalize EA as rigorously as medicine, law, engineering, accounting or even software engineering. Other camps are in favor of professionalization to the fullest extent. This debate is centered on whether to position EA as a legally recognized profession or as a generally accepted profession.

If EA is to be a generally accepted profession, it sets its own standards and governs itself at a minimum by the four E’s:

1. Education

2. Examination
3. Ethics
4. Experience

If EA is to be a legally recognized profession, it will also need to address:

- Government certification
- Licensed by State or by “Law”
- State board and society group requirements

Interactions between the above components are more complex than might be inferred. For example, the requirements for professional licensing can have a significant effect on the content of initial education. A professional society may manage the certification process or develop the code of ethics. Certification guidelines can influence the content of professional development materials.

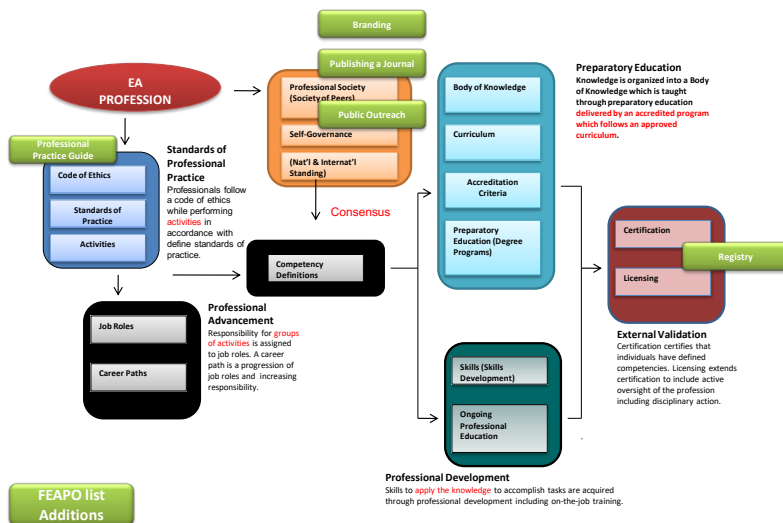
If either case (legally recognized or generally accepted) there are common traits shared by both types of professional status:

1. A body of knowledge rested on well-developed and widely accepted theoretical and practical bases.
2. A system for certifying that individuals possess such knowledge before practice.
3. A progressive system of certifying practitioners as they increase both their knowledge and experience in effective practice.
4. A code of ethics, with a commitment to use specialized knowledge for public good.
5. A professional society, with provisions for monitoring individual compliance.

The model that FEAPO has adopted for accelerating the recognition of EA as a true profession includes the elements needed for both: the full model reflects the elements needed for a legally recognized profession; omission of the licensing element is fine for a generally accepted profession.

Modeling a Profession

The combined model (below) of a profession (the basic model developed by the Computer Society and additions from CAEAP) defines the elements that comprise a formal profession. FEAPO has ratified this model as a grounding piece of its work.



The elements of the EA profession must address both the enterprise and technology. The required elements that the IEEE Computer Society has identified for an IT profession are described below. FEAPO intends to extend these elements to include the requisite enterprise areas.

Professional Societies

- Existence of professional organizations that support the advancement of the profession through such means as branding, publishing refereed papers, such as in a journal, and various forms of public outreach.
- Sufficiency and sustainability of employment in the profession.
- Active research efforts to advance the state of the profession's knowledge

Code of Ethics

- Common, applicable and useful to all EA sub-disciplines
- Specifies appropriate professional conduct to engage in the profession, including guidance for using specialized knowledge for the common good

Activities

- Specification, development, testing, of enterprise information systems, including enterprise personnel and processes, and the supporting technologies
- Application and management of such technologies, including development, operation, support and maintenance.

Standards of practice

- A Professional Practice Guide
- Application of current best practice in performing activities
- Currency in evolving technologies, methods
- International standards that specify techniques, methods, procedures and performance norms, consensually agreed by the profession

Competency Model

- Comprehensive competency model in place that describes knowledge, skills and levels of competence
- Ongoing maintenance regime for the competence model

Guide to the Body of Knowledge

- Clearly defined core body of knowledge that establishes the boundaries of the discipline
- Specialized bodies of knowledge beyond “core”
- Identification of supporting knowledge areas such as technical management and project management
- A description of the knowledge, methods, and practices that define the content of the profession.
- Consensual validation of the knowledge.
- A rational and scientific foundation for the knowledge.

Curriculum Model(s)

- Defined by recognized, authoritative bodies
- Provide a common basis for the establishment and improvement of educational programs.
- Aligned with relevant Competency Model(s)

Accreditation Criteria

- Aligned to certification requirements
- An accreditation system that assures the quality and suitability of the preparatory education.

Preparatory Education

- Readily available programs
- Complies with curriculum models and accreditation criteria
- Provides the education and training necessary to be employed in an entry level position in the profession.

Skills

- The ability to influence things; to take what you know and apply it so as to cause the desired effect to occur.

NOTE: A certain amount of knowledge is a prerequisite of skill; you can't be skillful without first being knowledgeable; however, you can easily be knowledgeable without being very skillful.

Professional Development

- Multiple sources of education and training to maintain and advance skills and knowledge
- Increasing experience necessary to keep current and advance in the profession.

Certification

- Standard and recognized certification regime
- Consistent with body of knowledge and competency model
- Mechanisms for competency certification that do not require compliance with education requirements (i.e., that enable recognition of competency acquired by other means)
- Defined means of re-certification/CPD & de-certification
- Includes professionalism aspects
- Includes demonstration of competency through practice (apprenticeship / experience)
- Mechanisms to support specialist competence areas
- Validation, by a community of peers, that an individual possesses the knowledge and competence of a professional.
- A public registry of certified professionals

Licensing

- Implemented by other authoritative bodies
- Supported by clear body of knowledge, certification standards, competency standards, code of ethics and a clear boundary of a licensed discipline
- A public registry of licensed professionals
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Job Roles

- Defined positions in an organization defined in terms of the responsibilities for the performance of specific activities assigned to the position and in terms of the competencies required to perform the activities
- Clear delineation of level of responsibility for the role

Career Paths

- A pre-defined series of job roles where experience in previous job roles is relevant to subsequent job roles.
- Paths for technical advancement in job roles based on competencies

The Roadmap

FEAPO member organizations are working together to accelerate the development of the various elements that are needed to support EA as a profession. Each of the individual members have developed artifacts that will contribute to the development of the elements, but lacked a means to prioritize their efforts to bring the disparate artifacts into a coherent whole. The method selected was to develop a roadmap for the professionalization of Enterprise Architecture.

The following diagram represents that roadmap -- a logical order of development of the elements of a profession, focusing on FEAPO's collaborative efforts to promote the professionalism of Enterprise Architecture.

The first step in the roadmap has already been accomplished: FEAPO has been established, and includes the key players from the field of organizations serving EA practitioners. These member organizations have individually established valuable artifacts and stores of knowledge supporting the practice of EA. FEAPO's challenge is to build on those in order to provide a single coherent "face" to the profession. Achieving this common face (and voice) necessarily involves collaboration in harmonizing the elements that already exist.

Figure 3 lays out a roadmap for this work. For example, before we can achieve agreement on what belongs in the Guide to the Enterprise Architecture Body of Knowledge (EABOK), we need to agree on the scope of EA—what's in and what's out. In order to agree on the knowledge areas in the Guide, we should have agreement on what skills and competencies a skillful EA practitioner demonstrates. These can be developed in tandem with (and will feed into) a list of topics for the BoK Guide. The skills/competencies and topics can then be sorted into Knowledge Areas (KAs) to be addressed in the Guide to the BoK.

Agreement on the Knowledge Areas facilitates agreement on how to use the FEAPO Member Organizations' (FMO) existing knowledge bases and artifacts. Some may be folded into or otherwise support the consensually agreed KAs in the Guide. In other cases, the Guide may cite an FMO's artifact(s) as a primary reference. In any case, the agreed KAs provide a basis for commonly accepted definitions of requisite competencies and thus provides a basis for developing a Model Curriculum. The Model Curriculum provides a basis for educational institutions to develop their own specific preparatory education courses.

Development of a competency model assembles the requisite competencies increasing levels of competency. This lays the basis for grouping competencies into prototypical job descriptions. The job descriptions can be used for laying out archetypes for possible career paths.

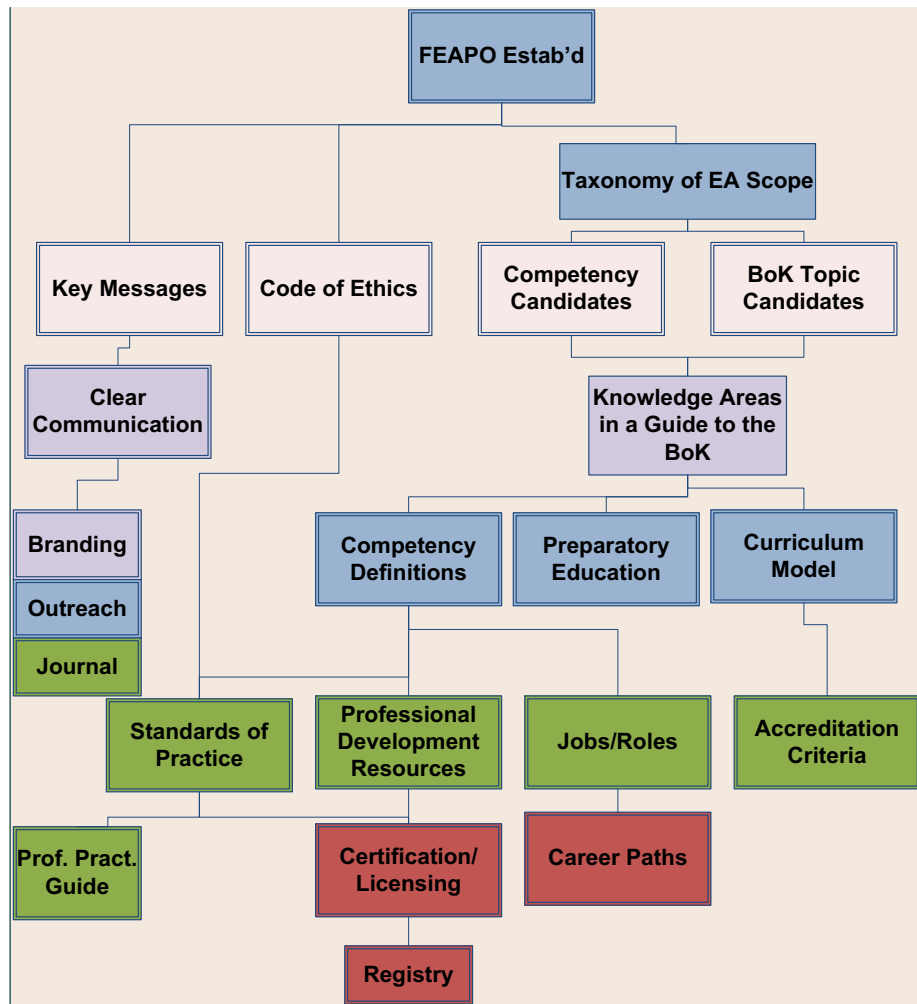


Figure 3: FEAPO's Roadmap for Development of the EA Profession

Globalization and the Profession

To attain recognition as a global profession, EA professionals must have clear ethical and competence standards that accommodate cultural differences in the regulation of professions. Given the ubiquitous role of EA, a wide range of individuals and organizations have a stake in the profession. Primary stakeholders are professionals and those who employ them. The practice and portability of professional

Commented [jw4]: Competency definitions are a subset of data which belong in a Data Dictionary, whereas the notion of a model is what we are attempting to convey with the suggestion of a taxonomy. If we are going to insert the word model, it might be best placed in the blue box since that is where the model will be defined (and it will likely have more than one taxonomy).

Commented [cw5]: Change Competency Definitions to Comp. Model; add an analysis of current career paths; add tech stds

skills are similar across diverse enterprise environments, as well as cultural and geographical boundaries.

In determining to accelerate EA's establishment as a global profession, FEAPO Member Organizations are working together to:

1. Instantiate the requisite elements of a profession described above
2. Promote the art and science of EA by uniting likeminded individuals in the major organizations of the EA ecosystem
3. Identify and establish a body of ethics, principles, and maturity measures that support the Enterprise Architecture profession and its practice.
4. Develop and certify a body of knowledge that properly represents the Enterprise Architecture Profession and its practice.
5. Improve the practice of Enterprise Architecture by providing consensually based artifacts, education and certification
6. Establish collaboration for skills development and professional development between professional societies
7. Establish collaboration avenues for professional education and model curricula developers to achieve accreditation.

Perhaps no human artifact is more important to the survival of the species than the organization of enterprises. Nearly everything humans do together is done in the context of some kind of enterprise – families, businesses, governments, social groups, educational institutions, military organizations, charities, and so on. Relatively small, static, or simple enterprises in general may not require an explicitly documented enterprise architecture. But most enterprises of consequence are large, dynamic, complicated, and important. Random, organic, component-focused growth and management are no longer viable options for enterprise management. The current global economic situation is strong evidence of this. Enterprises must be architected and managed by that architecture if they are to remain viable in the Information Age. The EA profession can make that happen.

Commented [cw6]: I loved this paragraph from ML as a coda to the doc, but Mark says it's been published before. How about if we quote the source?